A Literal Answer to Problem 1 in *Quantum Expanders and Geometry of Operator Spaces*

Automated mathematics run `fa_banach_001`

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Status. Candidate full answer to the statement as printed. The likely intended "for all sufficiently large N" version is not answered.

1 Source and printed problem

The source is G. Pisier, *Quantum Expanders and Geometry of Operator Spaces*, arXiv:1209.2059, Section 3, Problem 1, page 25 of the source PDF [1]. For a finite-dimensional operator space E, an integer $N \geq 1$, and $C > 1$, the number $k_E(N, C)$ is the least k for which there are linear maps $f_j : E \to M_N$ such that, for every $x \in M_N(E)$,

$$\sup_{1 \leq j \leq k} \|(\mathrm{Id} \otimes f_j)(x)\| \leq \|x\| \leq C \sup_{1 \leq j \leq k} \|(\mathrm{Id} \otimes f_j)(x)\|.$$

The printed question is:

> Let $C > 1$. Assume that a finite dimensional space E satisfies $k_E(N, C) \leq 1$ for all N. What does that imply on E? Is E exact with a control on its exactness constant?

2 Proof intuition

The universal quantifier contains the scalar level $N = 1$. At that level a single map into $M_1 = \mathbb{C}$ must recover the norm of every vector of E up to the factor C. Its lower estimate makes it injective, and an injective linear map from E into a one-dimensional space forces E itself to have dimension at most one. One-dimensional operator spaces have no residual matricial complexity: their matrix norms are exactly the scalar matrix norms, so they are completely isometric to $\mathbb{C}$.

3 Formal result

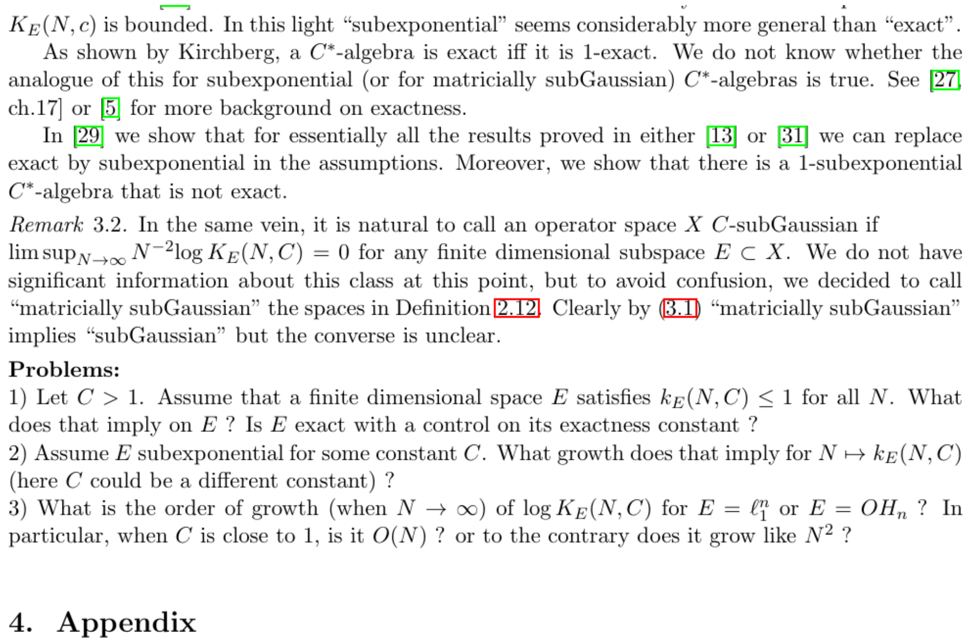
Theorem 1. *Let E be a finite-dimensional operator space and let $C > 1$. If*

$$k_E(N, C) \leq 1 \qquad \text{for every integer } N \geq 1,$$

then $\dim E \leq 1$. If $E \neq \{0\}$, then E is completely isometric to $\mathbb{C}$ and its exactness constant is

$$\mathrm{ex}(E) = 1.$$

In particular, the printed Problem 1 has an affirmative answer with optimal control independent of C.



$K_E(N, c)$ is bounded. In this light "subexponential" seems considerably more general than "exact".

As shown by Kirchberg, a C^*-algebra is exact iff it is 1-exact. We do not know whether the analogue of this for subexponential (or for matricially subGaussian) C^*-algebras is true. See [27, ch.17] or [5] for more background on exactness.

In [29] we show that for essentially all the results proved in either [13] or [31] we can replace exact by subexponential in the assumptions. Moreover, we show that there is a 1-subexponential C^*-algebra that is not exact.

Remark 3.2. In the same vein, it is natural to call an operator space X C-subGaussian if $\limsup_{N\to\infty} N^{-2}\log K_E(N, C) = 0$ for any finite dimensional subspace $E \subset X$. We do not have significant information about this class at this point, but to avoid confusion, we decided to call "matricially subGaussian" the spaces in Definition 2.12. Clearly by (3.1) "matricially subGaussian" implies "subGaussian" but the converse is unclear.

Problems:

1) Let $C > 1$. Assume that a finite dimensional space E satisfies $k_E(N, C) \le 1$ for all N. What does that imply on E ? Is E exact with a control on its exactness constant ?

2) Assume E subexponential for some constant C. What growth does that imply for $N \mapsto k_E(N, C)$ (here C could be a different constant) ?

3) What is the order of growth (when $N \to \infty$) of $\log K_E(N, C)$ for $E = \ell_1^n$ or $E = OH_n$? In particular, when C is close to 1, is it $O(N)$? or to the contrary does it grow like N^2 ?

4. Appendix

Figure 1: The complete problem list on source page 25; Problem 1 is the statement answered here.

Proof. Set $N = 1$. Since $M_1(E) = E$ and $M_1 = \mathbb{C}$, the hypothesis $k_E(1, C) \le 1$ supplies a linear map $f : E \to \mathbb{C}$ such that

$$|f(x)| \le \|x\|_E \le C|f(x)| \qquad (x \in E).$$

If $f(x) = 0$, the right-hand inequality gives $\|x\|_E = 0$, hence $x = 0$. Thus f is injective. Since its range lies in the one-dimensional vector space $\mathbb{C}$, it follows that $\dim E \le 1$.

Assume now that $E \ne \{0\}$. Realize E concretely as a subspace of $B(H)$ and choose $e \in E$ with $\|e\| = 1$. The map

$$u : \mathbb{C} \longrightarrow E, \qquad u(\lambda) = \lambda e,$$

is an isometry. At every matrix level m and for every scalar matrix $a = [\lambda_{pq}] \in M_m$,

$$\|u_m(a)\|_{M_m(E)} = \|[\lambda_{pq}e]\|_{B(\mathbb{C}^m\otimes H)} = \|a \otimes e\| = \|a\|_{M_m}\|e\| = \|a\|_{M_m}.$$

Therefore u is a complete isometry, so E is completely isometric to the subspace $\mathbb{C} = M_1$ of a matrix algebra. Hence $\mathrm{ex}(E) \le 1$. By the normalization of the exactness constant it is always at least 1, and consequently $\mathrm{ex}(E) = 1$. $\square$

4 Scope and formulation caveat

Immediately before introducing matricially subGaussian spaces, the source observes that a C-exact space has $k_E(N, c) = 1$ for all sufficiently large N whenever $c > C$. This makes it plausible that Problem 1 intended the eventual condition

$$k_E(N, C) \le 1 \quad \text{for all sufficiently large } N,$$

rather than the universal condition printed on page 25. The proof above uses $N = 1$ essentially and says nothing about that eventual version. The result should therefore be read as both a complete literal answer and a formulation correction.

5 Verification and novelty check

The proof uses only the source definition and elementary properties of concrete one-dimensional operator spaces. The source TeX and PDF were checked to confirm that Problem 1 literally says "for all N" and that N ranges over positive integers in the definition.

A bounded search was performed in the run's solution, attempt, proof-gap, and registry indexes; the local arXiv source corpus; the source's sequel and related exact-operator-space papers; and targeted web searches for the exact notation $k_E(N, C)$, matricially subGaussian spaces, and later discussions of the problem. No explicit published observation of the $N = 1$ collapse was found. Novelty confidence is high for the literal formulation correction but low for any substantive intended problem, since the eventual-N question is not resolved here.

Human review should focus on whether an implicit convention in the paper was meant to exclude $N = 1$. If so, the packet remains correct only as a diagnosis of the printed wording.

References

[1] G. Pisier, *Quantum Expanders and Geometry of Operator Spaces*, J. Eur. Math. Soc. 16 (2014), 1183–1219; arXiv:1209.2059.